

Geometric Transformations

A geometric transformation changes the position, orientation, size, or shape of an object in an image or signal without changing the pixel values themselves.

Why Geometric Transformations are Needed

- Image alignment (registration)
- Rotation and scaling of images
- Object tracking
- Medical image analysis
- Satellite and aerial image correction
- Computer graphics and animation

Basic Idea (Pixel Mapping)

A transformation maps:

Two ways to map:

1. **Forward mapping** – move pixels forward
2. **Backward mapping** – find source pixel for each output pixel (preferred)

Types of Geometric Transformations

A. Rigid Transformations

- Translation
- Rotation
- Preserve shape and size

B. Similarity Transformations

- Translation
- Rotation
- Uniform scaling

C. Affine Transformations

- Translation
- Rotation

- Scaling
- Shearing
- Preserve parallel lines

D. Projective Transformations

- Perspective effects
- Do NOT preserve parallelism

Mathematical Representation (Homogeneous Coordinates)

To combine transformations easily, we use **homogeneous coordinates**:

1. Translation

Meaning

Move an object **left/right/up/down**

Equation

Matrix form

Numerical Example

1. Point:

Translate by:

✓ New point = **(6, 5)**

2. Scaling

Meaning

Change size of object

Equation

Matrix

Point:

Scale factors:

✓ New point = **(6, 2)**

3. Rotation

Meaning

Rotate around origin (counter-clockwise)

Equation

Matrix

Point:

Rotate by **90°**

✓ New point = **(0, 1)**

Classification of Geometric Transformations

Geometric transformations can be broadly classified into:

1. **Parametric (Global) Transformations**
2. **Non-parametric (Local) Transformations**
 - Mesh-based warping
 - Spline-based warping

This classification depends on whether the transformation is governed by a **small fixed set of parameters** or by **many local control points**.

Parametric Transformations

A geometric transformation in which the mapping between input and output coordinates is defined by a finite set of parameters that globally control image deformation.

Concept

A **parametric transformation** is a geometric transformation in which the mapping between input and output coordinates is defined using a **finite number of parameters**. These parameters globally control the deformation of the entire image.

All pixels undergo the **same transformation rule**.

Mathematical Representation

General parametric form:

where:

- = original coordinates
- = transformed coordinates
- = transformation parameters

Common Parametric Transformations

(a) Translation (2 parameters)

What are the parameters?

- $\rightarrow$ horizontal shift
- $\rightarrow$ vertical shift

These two values completely define the translation.

(b) Scaling (2 parameters)

What are the parameters?

- $\rightarrow$ scaling along x-direction
- $\rightarrow$ scaling along y-direction

These two numbers define how much stretching/shrinking happens.

(c) Rotation (1 parameter)

What is the parameter?

- $\rightarrow$ rotation angle

Just **one number** decides everything.

(d) Affine Transformation (6 parameters)

Parameters:

Parameter	Meaning
(a)	scaling / rotation
(b)	shear / rotation
(c)	shear / rotation
(d)	scaling / rotation
(t_x)	translation in x
(t_y)	translation in y

Affine transformations preserve:

- Straight lines
- Parallelism
but not angles or lengths.

Homogeneous Coordinate Representation

To combine multiple parametric transformations, homogeneous coordinates are used:

Properties of Parametric Transformations

- ✓ Global transformation
- ✓ Few parameters
- ✓ Computationally efficient
- ✓ Easy to estimate and invert

Applications

- Image registration
- Camera motion correction
- Medical image alignment
- Object tracking
- Document image correction

Mesh-Based Warping

A non-parametric image warping technique that divides the image into a mesh and applies local transformations using displaced control points.

Concept

Mesh-based warping is a **local, non-parametric geometric transformation** in which the image is divided into a grid (mesh). Each mesh vertex acts as a **control point**. By moving these control points, local deformations are created.

Different regions of the image can deform differently.

Mesh Construction

1. Divide the image into:
 - Rectangular mesh, or
 - Triangular mesh
2. Define control points at mesh vertices
3. Specify new positions for control points

Warping Process (Step-by-Step)

1. **Mesh generation** over original image
2. **Control point displacement**
3. **Local transformation** of each mesh cell
4. **Interpolation** of pixels inside each cell

Mathematical Formulation (Conceptual)

Let:

- = original mesh vertex
- = displaced mesh vertex

For a point inside a mesh element:

where:

- are interpolation weights
- for rectangular mesh
- for triangular mesh

Interpolation Methods

- **Bilinear interpolation** (rectangular mesh)
- **Barycentric interpolation** (triangular mesh)

Properties of Mesh-Based Warping

- ✓ Local deformation capability
- ✓ Highly flexible
- ✓ Suitable for non-rigid transformations

- ✗ Higher computational cost
- ✗ Requires many control points
- ✗ More complex implementation

Applications

- Face morphing and animation
- Medical image deformation modeling
- Image editing and special effects
- Shape matching
- Non-rigid image registration

Example 1:

Point: $(x,y)=(2,3)$ perform Translation.

Formula

Parameters

Step-by-step

Output point

Example 2:

Point: $(x,y)=(2,3)$, perform Scaling.

Formula

Parameters

Step-by-step

Output point

Example 3:

Point: $(x,y)=(2,3)$, perform Rotation.

Formula

Parameter

Step-by-step

Output point

Example 4:

Point: $(x,y)=(2,3)$, perform Affine Transformation.

Formula

Parameters (given)

Step-by-step

First matrix multiplication:

Output point